Linear Algebra

Revision companion

Use this after the guided lessons: cover each worked answer, recall the rule, then solve the example on paper. Every module includes its key formulas, conditions and common traps. Full explanations and runnable examples are in the companion HTML notes.

Scope: Every named Section 2 topic, plus explicitly identified applications such as QR, Cholesky, pseudoinverses and PCA.

Official GATE DA 2027 syllabus ↗

01 Vectors & geometric intuition

02 Spaces, subspaces & bases

03 Matrix operations & blocks

04 Special matrices

05 Systems & Gaussian elimination

06 Rank, nullity & fundamental spaces

07 Determinants & invertibility

08 Orthogonality & least squares

09 Eigenvalues & eigenvectors

10 Quadratic forms & curvature

11 LU, pivoting & stable solves

12 SVD, pseudoinverse & low-rank models

01 / Vectors & geometric intuition

Coordinates, linear combinations, lengths, angles and linear maps

Scalars, vectors and linear combinations

A scalar is one number; a vector is an object that can be added and scaled. In R^2 or R^3, draw a vector as an arrow from the origin. Its coordinates describe it relative to an ordered basis. A point and a displacement can have the same coordinate list, but their roles differ.

$$a\,u + b\,v = (a\,u_1+b\,v_1, \ldots, a\,u_n+b\,v_n)^T$$

Work it through: Let u=(1,2), v=(2,−1). Find 3u+2v. 3u=(3,6) and 2v=(4,−2). Add components: 3u+2v=(7,4). Conversely, solve a+2b=7 and 2a−b=4 to recover a=3,b=2.

Watch the trap: A line not passing through the origin is an affine set, not a vector subspace.

Recall task: For u=(1,2),v=(2,−1), compute −u+3v and explain the reversed arrow.

Worked answer: −u+3v=(5,−5). The coefficient −1 reverses u.

Dot products, norms and angles

For real coordinate vectors, u^Tv is the sum of products of corresponding components. The Euclidean norm is the square root of u^Tu. It measures length and is zero only for the zero vector. The 1-norm sums absolute values; the infinity norm is the largest absolute component.

$$u^Tv = \Sigma_i u_i v_i \cdot ||u||_2 = \sqrt{(u^Tu)} \cdot \cos\theta = u^Tv/(||u||_2||v||_2)$$

Work it through: Take u=(3,4),v=(4,−3). $u^Tv=12-12=0$, so the vectors are perpendicular. Both norms are 5. Their normalized versions are (0.6,0.8) and (0.8,−0.6). For w=(1,1), $||w||_1=2$, $||w||_2=\sqrt{2}$ and $||w||_\infty=1$.

Watch the trap: Orthogonal means dot product zero. Orthonormal additionally requires unit length.

Recall task: Find the norm and unit vector for (−2,1,2).

Worked answer: Norm 3; unit vector (−2/3,1/3,2/3).

Linear transformations and the column picture

A linear map T preserves addition and scaling: T(a u+b v)=aT(u)+bT(v). Consequently T(0)=0. A map can preserve zero without being linear, so use both defining properties. Translation by a nonzero vector is affine rather than linear.

$$A=[T(e_1)\ \ldots\ T(e_n)] \Rightarrow Ax=\Sigma_i x_i T(e_i) \cdot (AB)x=A(Bx)$$

Work it through: Let A=[[1,1],[0,1]]. Its columns are (1,0) and (1,1). For x=(2,3), Ax=2(1,0)+3(1,1)=(5,3). Horizontal coordinates shift by the vertical coordinate: this is a shear. The origin is fixed and det(A)=1, so area is preserved.

Watch the trap: The rightmost matrix acts first when transformations act on column vectors.

Recall task: Apply a 90° counterclockwise rotation to (2,1).

Worked answer: Using [[0,−1],[1,0]] gives (−1,2).

02 / Spaces, subspaces & bases

Axioms, closure tests, independence, dimension and coordinates

Vector spaces and the subspace test

A vector space is a set with addition and scalar multiplication satisfying the vector-space axioms: associativity and commutativity of addition, a zero vector, additive inverses, the two distributive laws, associative scalar action and multiplication by scalar 1. Scalars come from a specified field, usually R in this guide.

$W \le V \Leftrightarrow W \neq \emptyset$ and $a\,u+b\,v \in W$ for every $u,v \in W$ and $a,b \in R$

Work it through: Let $W=\{(x,y,z): x+2y-z=0\}$. Solve for x: $x=-2y+z$. Every vector is $y(-2,1,0)+z(1,0,1)$. W is a plane through zero, spanned by those two independent vectors.

Watch the trap: “Degree exactly d” is not a polynomial vector space: zero is excluded and cancellation can lower degree.

Recall task: Show that $\{(x,y): x \ge 0\}$ is not a subspace.

Worked answer: (1,0) belongs, but scalar multiplication by -1 gives $(-1,0)$, which does not.

Dependence, independence and redundancy

Vectors $v_1,\ldots,v_k$ are independent when $c_1v_1+\ldots+c_kv_k=0$ has only the all-zero coefficient solution. Dependence means a nontrivial coefficient choice gives zero. In a dependent finite set, at least one vector can be written as a combination of the others.

$[v_1 \ \ldots \ v_k]c=0$ has only $c=0 \Leftrightarrow \{v_1,\ldots,v_k\}$ is independent

Work it through: Let $v_1=(1,0), v_2=(0,1), v_3=(1,1)$. $v_1+v_2-v_3=0$ is a nontrivial relation. All three pairs are nonparallel, but the three-vector set is dependent. Keep v_1,v_2 to preserve the same span R^2 without redundancy.

Watch the trap: Independent rows and independent columns have different counts; for a rectangular matrix, say which independence you mean.

Recall task: Find a dependence relation for (1,2),(2,4).

Worked answer: $2(1,2)-(2,4)=0$, so the coefficients $(2,-1)$ are nontrivial.

Basis, dimension and change of coordinates

A basis is an independent spanning set. Every vector in the space has a unique coordinate representation in that basis: spanning gives existence, independence gives uniqueness. The number of basis vectors is the dimension; all bases of the same finite-dimensional space have the same size.

$x=B[x]_B \cdot [x]_C=C^{-1}B[x]_B \cdot \dim(U+W)=\dim U+\dim W-\dim(U\cap W)$

Work it through: Choose $B=[(1,1),(1,-1)]$ and $x=(4,2)$. The coordinate equations are $c_1+c_2=4$ and $c_1-c_2=2$. Therefore $c_1=3, c_2=1$, so $[x]_B=(3,1)$. The standard coordinates (4,2) and B-coordinates (3,1) describe the same vector.

Watch the trap: An independent set need not span the whole ambient space; specify the space for which it is a basis.

Recall task: Find the coordinates of (2,4) in the same basis.

Worked answer: $c_1+c_2=2$ and $c_1-c_2=4$ give $(3,-1)$.

03 / Matrix operations & blocks

Dimensions, products, transpose, inverse and partitioned matrices

Matrix arithmetic without shape mistakes

An m×n matrix has m rows and n columns. Addition requires the same shape and is componentwise. Scalar multiplication scales every entry. For A of shape m×n and B of shape n×p, AB has shape m×p; the inner dimensions must agree.

$$(AB)_{ij}=\Sigma_k A_{ik}B_{kj} \cdot (m\times n)(n\times p)\to m\times p$$

Work it through: A=[[1,2],[0,1]], B=[[2,0],[1,3]]. AB first row is (1·2+2·1,1·0+2·3)=(4,6). AB second row is (1,3), so AB=[[4,6],[1,3]]. BA=[[2,4],[1,5]], confirming that order matters.

Watch the trap: AB=AC does not imply B=C unless cancellation by A is justified, for example A is invertible.

Recall task: Compute AB and BA for A=[[1,1],[0,1]], B=[[1,0],[1,1]].

Worked answer: AB=[[2,1],[1,1]], BA=[[1,1],[1,2]].

Transpose, inverse and useful identities

The transpose exchanges rows and columns: $(A^T)_{ij}=A_{ji}$. It reverses a product: $(AB)^T=B^TA^T$. The trace is the sum of diagonal entries of a square matrix. For compatible products, tr(AB)=tr(BA); this does not mean AB=BA.

$$(AB)^T=B^TA^T \cdot (AB)^{-1}=B^{-1}A^{-1} \cdot A^{-1}=(1/(ad-bc))[[d,-b],[-c,a]]$$

Work it through: A=[[2,1],[1,1]] has determinant 1. Its inverse is [[1,−1],[−1,2]]. For b=(3,2), $x=A^{-1}b=(1,1)$. Verify by multiplying A(1,1)=(3,2), rather than trusting the formula alone.

Watch the trap: A matrix with a zero diagonal entry can still be invertible. [[0,1],[1,0]] is a counterexample.

Recall task: Invert [[1,2],[3,4]] and verify its product with the original.

Worked answer: The inverse is [[−2,1],[1.5,−0.5]]. Both products equal I.

Partitioned matrices and Schur complements

A partitioned or block matrix groups rows and columns into submatrices. The underlying matrix is unchanged. Block addition and multiplication use the same rules as scalar entries, but each block product must have compatible dimensions and its order must be preserved.

$$S=D-CA^{-1}B \cdot \det([[A,B],[C,D]])=\det(A)\det(S)\ [A\ \text{invertible}]$$

Work it through: For M=[[2,1],[1,3]], use scalar blocks A=2,B=C=1,D=3. S=3−1·(1/2)·1=2.5. det(M)=2·2.5=5, matching 2·3−1·1. For block sizes larger than one, the multiplication order in $CA^{-1}B$ becomes essential.

Watch the trap: A partition is a grouping device; it does not make matrix blocks commute.

Recall task: Use a Schur complement to compute det([[4,2],[2,3]]).

Worked answer: S=3−2(1/4)2=2; determinant=4·2=8.

04 / Special matrices

Symmetry, orthogonality, idempotence and projection operators

Symmetric, skew-symmetric and triangular matrices

A real square matrix is symmetric if $A^T=A$ and skew-symmetric if $A^T=-A$. A skew-symmetric real matrix has zero diagonal, because $a_{ii}=-a_{ii}$. Every real square matrix splits uniquely into a symmetric and a skew-symmetric part.

$A=S+K$, $S=(A+A^T)/2$, $K=(A-A^T)/2$ · $x^TKx=0$

Work it through: For A=[[1,4],[2,3]], S=[[1,3],[3,3]]. K=[[0,1],[−1,0]], and S+K=A. For x=(2,1), $x^TKx=2-2=0$. Therefore $x^TAx=x^TSx=19$.

Watch the trap: A zero diagonal does not imply skew-symmetry; the paired off-diagonal entries must also be negatives.

Recall task: Split [[2,5],[1,4]] into symmetric and skew-symmetric parts.

Worked answer: S=[[2,3],[3,4]], K=[[0,2],[−2,0]].

Orthogonal matrices preserve geometry

A real square matrix Q is orthogonal when $Q^TQ=I$. Its columns form an orthonormal basis, and its rows do too. $Q^{-1}=Q^T$. A rectangular matrix may have orthonormal columns, but is not an orthogonal square matrix under this convention.

$Q^TQ=I \Rightarrow Q^{-1}=Q^T$, $||Qx||_2=||x||_2$, $\det Q=\pm1$

Work it through: Q=[[0,−1],[1,0]] rotates by 90°. $Q^TQ=I$ and det Q=1. For x=(3,4), Qx=(−4,3). Both vectors have norm 5. A reflection diag(1,−1) also preserves length but has determinant −1.

Watch the trap: If U is tall with orthonormal columns, $U^TU=I$ does not imply $UU^T=I$.

Recall task: Verify that diag(1,−1) is orthogonal and reflects (2,3).

Worked answer: Its transpose times itself is I; the output is (2,−3), with the same norm.

Idempotent matrices and projection matrices

An idempotent matrix satisfies $P^2=P$: applying it twice has the same effect as applying it once. Its image consists of fixed vectors, and its kernel consists of vectors sent to zero. Every vector x decomposes as Px+(I−P)x into image and kernel components.

$P^2=P$; orthogonal projection also has $P^T=P$. $P_{line}=uu^T/(u^Tu)$, $u\neq0$

Work it through: Project onto the line spanned by u=(1,1). P=(1/2)[[1,1],[1,1]]. For x=(3,1), Px=(2,2), and residual (1,−1) is perpendicular to u. $P^2=P$, tr(P)=1 and rank(P)=1.

Watch the trap: Idempotence alone does not imply orthogonal projection. [[1,1],[0,0]] is idempotent but not symmetric.

Recall task: Build the projector onto (1,2) and project (3,0).

Worked answer: P=(1/5)[[1,2],[2,4]], giving (0.6,1.2).

05 / Systems & Gaussian elimination

Augmented matrices, pivots, consistency and parametric solutions

Translate equations into Ax=b

A system of m linear equations in n unknowns is Ax=b, where each row stores one equation’s coefficients. A solution makes all equations true simultaneously. Geometrically in R^2, two nonparallel lines intersect once, parallel distinct lines never meet, and coincident lines have infinitely many common points.

$$Ax=b \text{ is consistent} \Leftrightarrow b \in \text{Col}(A) \Leftrightarrow \text{rank}(A)=\text{rank}([A|b])$$

Work it through: Solve x+y=3 and x−y=1. Adding equations gives 2x=4, so x=2. Substitute to obtain y=1. The matrix columns (1,1) and (1,−1) combine as 2(1,1)+(1,−1)=(3,1).

Watch the trap: A homogeneous system is always consistent, even when A is singular.

Recall task: Compare x+y=2, 2x+2y=4 with x+y=2, 2x+2y=5.

Worked answer: The first has x=2−t,y=t for any real t. The second reduces to 0=1 and is inconsistent.

Gaussian elimination, echelon form and pivoting

Gaussian elimination uses a pivot to remove entries below it, producing row echelon form. Leading nonzero entries move right as you move down rows; zero rows go to the bottom. Back substitution then determines pivot variables from the last nonzero row upward.

$$R_i \leftarrow R_i - (a_{ik}/a_{kk})R_k \text{ [nonzero pivot } a_{kk}]$$

Work it through: For x+2y=5 and 3x+4y=11, start with [[1,2|5],[3,4|11]]. $R_2 \leftarrow R_2-3R_1$ gives [[1,2|5],[0,−2|−4]]. Back substitution: y=2, then x=1. The row multiplier 3 will become an entry in L in LU factorization.

Watch the trap: A zero pivot at the current position does not prove singularity; a row exchange may repair it.

Recall task: Solve 2y=4 and x+y=3, explicitly swapping the initial zero-pivot row.

Worked answer: Swap equations first; y=2 then x=1.

Free variables and parameter-dependent systems

A row [0 … 0|c] with c≠0 is a contradiction. If no such row exists, the system is consistent. A consistent system with rank r and n unknowns has n−r free parameters. If r=n the solution is unique; otherwise infinitely many solutions exist over R.

$$\text{Consistent: } x=x_p+N\,t, \text{ where columns of } N \text{ span } \text{Null}(A); \text{ number of parameters}=n-\text{rank}(A)$$

Work it through: Consider x+y=2 and x+a y=3. Subtract the first equation. (a−1)y=1. If a≠1, y=1/(a−1), x=2−y. If a=1, the equations imply 0=1, so no solution. If the second right side were 2 instead, a=1 would produce infinitely many solutions.

Watch the trap: “Determinant zero” does not distinguish zero solutions from infinitely many solutions.

Recall task: Solve x+2y+z=3 and y−z=1 in parametric form.

Worked answer: Set z=t. Then y=1+t and x=1−3t, so x-vector=(1,1,0)+t(−3,1,1).

06 / Rank, nullity & fundamental spaces

Pivot counting, kernels, column spaces and dimension identities

Rank-nullity and the four fundamental spaces

The column space of an m×n matrix is a subspace of R^m; the row space is a subspace of R^n. Both have dimension rank(A)=r. The null space consists of x in R^n with Ax=0. The left null space consists of y in R^m with $A^Ty=0$.

$$A \in R^{m \times n},\ \text{rank } A=r \Rightarrow \text{nullity } A=n-r,\ \text{nullity } A^T=m-r$$

Work it through: A=[[1,2,3],[2,4,6]] has rank 1. Its nullity is 3−1=2 and left nullity is 2−1=1. A column-space basis is {(1,2)}. A row-space basis is {(1,2,3)}. Null(A) has basis {(-2,1,0),(-3,0,1)}.

Watch the trap: Use pivot columns of the original matrix for a column-space basis, not the altered columns of RREF.

Recall task: For a 4×7 matrix of rank 3, find nullity and left nullity.

Worked answer: Nullity 4; left nullity 1.

Rank inequalities and invertible transformations

Multiplication by an invertible matrix on either side preserves rank. Row operations are multiplication by invertible elementary matrices, explaining rank preservation. Transposition also preserves rank. Multiplication by a singular matrix can reduce rank but cannot increase it beyond the factors' ranks.

$$\text{rank } A+\text{rank } B-n \le \text{rank}(AB) \le \min(\text{rank } A,\text{rank } B) \cdot \text{rank}(A^TA)=\text{rank } A$$

Work it through: A is 4×5 with rank 4; B is 5×3 with rank 3. The product is 4×3. Its rank is at most 3. The lower bound is 4+3−5=2. So rank(AB) can only be 2 or 3; factor ranks alone do not choose between them.

Watch the trap: rank(AB)=rank(A)rank(B) is not a valid identity.

Recall task: For 3×3 matrices A,B each of rank 2, give possible rank bounds on AB.

Worked answer: 1≤rank(AB)≤2.

Orthogonal complements and solvability certificates

The orthogonal complement W⊥ contains vectors perpendicular to every vector of W. In R^n, dim W+dim W⊥=n and W∩W⊥={0}. Every vector has a unique orthogonal decomposition into a component in W and one in W⊥.

$$Ax=b \text{ solvable} \Leftrightarrow y^Tb=0 \text{ for every } y \text{ with } A^Ty=0$$

Work it through: A=[[1,1],[2,2]] and y=(−2,1). Then $A^Ty=0$. For b=(1,2), $y^Tb=0$, and the system is consistent. For b=(1,3), $y^Tb=1$, proving inconsistency. The certificate is exactly "second equation minus twice the first gives 0=1".

Watch the trap: Orthogonality to one left-null vector suffices only if it spans the entire left null space; otherwise test a basis.

Recall task: For A=[[1],[1],[1]], characterize all b for which Ax=b is solvable.

Worked answer: All three entries of b must be equal; b must lie on the line spanned by (1,1,1).

07 / Determinants & invertibility

Area, row-operation rules, cofactors and parameter cases

Determinants as signed volume scaling

The determinant of a square matrix is its signed volume scale factor. In 2D, $|\det A|$ is the area of the parallelogram formed by its columns. A negative determinant reverses orientation. A zero determinant means the columns collapse volume into a lower-dimensional set.

$$\det [[a,b],[c,d]]=ad-bc \cdot \text{area scale}=|\det A| \cdot \det A\neq 0 \Leftrightarrow A \text{ invertible}$$

Work it through: $A=[[2,1],[0,3]]$ sends the unit square to a parallelogram. Its area is $|2\cdot 3-0\cdot 1|=6$. Multiplying by a reflection changes the determinant sign but not the area magnitude. If the second column were twice the first, the parallelogram would collapse and the determinant would be zero.

Watch the trap: $\det(A+B)$ is not generally $\det(A)+\det(B)$.

Recall task: Find determinant and orientation effect of $[[1,3],[2,1]]$.

Worked answer: $1-6=-5$: area scales by 5 and orientation reverses.

Efficient determinant computation

A row swap multiplies the determinant by -1. Multiplying one row by c multiplies it by c. Adding a multiple of one row to another leaves the determinant unchanged. Track these changes while reducing to triangular form, then multiply the diagonal.

$$\text{Row add: unchanged; row swap: } \times(-1); \text{ one-row scale c: } \times c; \det(cA)=c^n \det A$$

Work it through: $A=[[1,2,0],[2,5,1],[0,1,3]]$. $R_2 \leftarrow R_2-2R_1$ gives second row $[0,1,1]$. $R_3 \leftarrow R_3-R_2$ gives $[0,0,2]$. No swaps or row scalings occurred, so $\det A=1\cdot 1\cdot 2=2$.

Watch the trap: For $n\times n$ A, multiplying every entry by 2 multiplies the determinant by 2^n, not merely 2.

Recall task: If a 3×3 A has determinant 5, find $\det(2A^T)$.

Worked answer: $2^3 \det(A^T)=8\cdot 5=40$.

Determinant parameters and structural shortcuts

For parameter-dependent matrices, compute a determinant polynomial and factor it to locate singular cases. Outside those roots the matrix is invertible. At a root, return to rank or the augmented system to decide the remaining properties.

$$\det(I+uv^T)=1+v^Tu \cdot \det [[A,B],[0,D]]=\det A \cdot \det D$$

Work it through: $A(t)=[[t,1],[1,t]]$ has determinant $t^2-1=(t-1)(t+1)$. It is singular exactly at $t=1$ or $t=-1$. At $t=1$, both rows are identical and rank is 1. For $u=(1,2), v=(3,4)$, $\det(I+uv^T)=1+3+8=12$.

Watch the trap: A determinant root identifies singularity, not the entire solution behavior of $Ax=b$.

Recall task: Determine the singular parameter values of $[[a,2],[2,a]]$.

Worked answer: $a^2-4=0$, so $a=\pm 2$.

08 / Orthogonality & least squares

Projections, Gram-Schmidt, QR and the best-fit interpretation

Project a vector onto a line or subspace

The orthogonal projection of b onto a nonzero vector a is $p=a(a^Tb)/(a^Ta)$. The scalar coefficient is chosen so that residual $b-p$ is perpendicular to a. The projection p lies on the line, and the residual measures the shortest displacement from b to that line.

$p=a(a^Tb)/(a^Ta)$ · $P=A(A^TA)^{-1}A^T$ when A has independent columns

Work it through: Let a=(1,2), b=(3,1). $a^Tb=5$ and $a^Ta=5$, so p=(1,2). The residual is (2,−1), with $a^Tr=2-2=0$. The squared lengths are 10=5+5.

Watch the trap: The projection coefficient is a^Tb only when a is already a unit vector.

Recall task: Project (2,0) onto span(1,1), and find the squared error.

Worked answer: Projection (1,1), residual (1,−1), squared error 2.

Gram-Schmidt and QR factorization

Gram-Schmidt turns independent vectors into an orthonormal basis for the same span. Normalize the first vector. For the next vector, subtract its projection on every earlier orthonormal direction, then normalize what remains. A zero remainder signals dependence.

$v_j=a_j-\Sigma_{i<j}(q_i^Ta_j)q_i$; $q_j=v_j/||v_j||$; $A=QR$

Work it through: $a_1=(1,1)$, $a_2=(1,0)$. Then $q_1=(1,1)/\sqrt{2}$. $q_1^Ta_2=1/\sqrt{2}$; subtract (1/2,1/2) from a_2. The remainder is (1/2,−1/2); normalize to $q_2=(1,-1)/\sqrt{2}$. $R=[[\sqrt{2},1/\sqrt{2}],[0,1/\sqrt{2}]]$ for this sign convention.

Watch the trap: Subtract projections on the orthonormal vectors, not blindly on the original unnormalized columns.

Recall task: Orthogonalize (1,0,1) and (1,1,0); verify the remainder is orthogonal to the first.

Worked answer: Projection coefficient 1/2; remainder (1/2,1,−1/2), whose dot product with (1,0,1) is zero.

Least squares from geometry to regression

When $Ax=b$ is inconsistent, least squares chooses $\hat{x}$ minimizing $||Ax-b||_2^2$. The fitted vector $A\hat{x}$ is the orthogonal projection of b onto Col(A). Its residual is perpendicular to every column, yielding $A^TA\,\hat{x}=A^Tb$, the normal equations.

$\min_x ||Ax-b||^2 \Rightarrow A^T(A\hat{x}-b)=0$; full column rank $\Rightarrow \hat{x}=(A^TA)^{-1}A^Tb$

Work it through: Fit a constant c to b=(1,2,6); A is a column of three ones. Normal equation: 3c=1+2+6=9, giving c=3. Fitted values are (3,3,3), residual (−2,−1,3). The residual sums to zero, so it is orthogonal to the column of ones.

Watch the trap: An inconsistent system can have a unique least-squares minimizer even though it has no exact solution.

Recall task: Fit a line with intercept through noisy observations t=[0,1,2], y=[1,2,2].

Worked answer: $\beta_0=7/6$ and $\beta_1=1/2$; fitted values are 7/6,5/3,13/6.

09 / Eigenvalues & eigenvectors

Invariant directions, multiplicities, diagonalization and spectral reasoning

Find eigenvalues and eigenspaces

An eigenvector $v \neq 0$ satisfies $Av=\lambda v$: its direction is unchanged apart from stretching or reversal. The zero vector is never an eigenvector. Eigenvalues are defined for square matrices. An eigenvalue may be zero, corresponding to a nonzero vector in the null space.

$$Av=\lambda v,\ v\neq 0 \Leftrightarrow \det(A-\lambda I)=0 \text{ and } v\in \text{Null}(A-\lambda I)\{0\}$$

Work it through: $A=[[2,1],[1,2]]$. $\det(A-\lambda I)=(2-\lambda)^2-1=(\lambda-1)(\lambda-3)$. For $\lambda=3$, $v=(1,1)$; for $\lambda=1$, $v=(1,-1)$. The sum $3+1=4$ equals trace; product 3 equals determinant. Normalize these vectors if an orthonormal eigenbasis is needed.

Watch the trap: A repeated eigenvalue does not determine how many independent eigenvectors exist.

Recall task: Find eigenvalues of $[[4,1],[0,2]]$ and one vector for each.

Worked answer: $\lambda=4$ has $(1,0)$; $\lambda=2$ has $(1,-2)$.

Multiplicity and diagonalization

Algebraic multiplicity counts how often an eigenvalue occurs as a characteristic-polynomial root. Geometric multiplicity is the dimension of its eigenspace, equal to $\text{nullity}(A-\lambda I)$. For each eigenvalue, geometric multiplicity lies between 1 and its algebraic multiplicity.

$$A=S\Lambda S^{-1} \Leftrightarrow n \text{ independent eigenvectors; symmetric real } A \Rightarrow A=Q\Lambda Q^{T}$$

Work it through: For $J=[[1,1],[0,1]]$, $J-I=[[0,1],[0,0]]$. The null equation forces $v_2=0$, leaving only $\text{span}(1,0)$. Algebraic multiplicity is 2 but geometric multiplicity is 1. J is not diagonalizable, unlike I_2, whose eigenspace has dimension 2.

Watch the trap: A repeated eigenvalue neither guarantees nor rules out diagonalizability.

Recall task: Compare eigenspace dimensions for I_3 and $[[2,1],[0,2]]$.

Worked answer: I_3 has eigenvalue 1 with eigenspace dimension 3. The 2×2 Jordan block has eigenvalue 2 with eigenspace dimension 1.

Powers, similarity and eigenvalue shortcuts

If $Av=\lambda v$, then $A^k v=\lambda^k v$ for nonnegative integers k. More generally $p(A)v=p(\lambda)v$ for a polynomial p. If A is invertible, $A^{-1}v=(1/\lambda)v$. These facts transform eigenvalues without recalculating a full determinant.

$$p(A)v=p(\lambda)v \cdot A^2-\text{tr}(A)A+\det(A)I=0 \text{ for } 2\times2 A$$

Work it through: For $A=[[2,1],[1,2]]$, trace=4 and determinant=3. Cayley-Hamilton gives $A^2=4A-3I$. Multiplying once more gives $A^3=4A^2-3A=13A-12I$. The eigenvalues of A^3 are $3^3=27$ and $1^3=1$.

Watch the trap: Row reduction preserves rank but does not generally preserve eigenvalues; similarity does.

Recall task: If eigenvalues of invertible A are 2 and -1, find those of A^2+3I and A^{-1}.

Worked answer: A^2+3I: 7 and 4. A^{-1}: 1/2 and -1.

10 / Quadratic forms & curvature

Definiteness, principal axes, Rayleigh quotients and quadratic optimization

Reading and classifying a quadratic form

A real quadratic form is $q(x)=x^TAx$. Only the symmetric part $S=(A+A^T)/2$ matters: the skew part contributes zero. Write each cross term across two symmetric entries. Thus $2x^2+6xy+5y^2$ corresponds to $[[2,3],[3,5]]$, not $[[2,6],[6,5]]$. A linear or constant term belongs to a quadratic function but not to its homogeneous quadratic form.

$q(x)=x^TSx$, $S=S^T$ · SPD ⇔ every $\lambda_i>0$ · PSD ⇔ every $\lambda_i\geq 0$

Work it through: For $q=2x^2+2xy+2y^2$, $A=[[2,1],[1,2]]$. Its eigenvalues are 3 and 1, both positive. Alternatively, leading minors are 2 and 3, both positive. Thus $q>0$ away from the origin; its level sets are ellipses.

Watch the trap: Positive diagonal entries or a positive determinant alone do not establish positive definiteness.

Recall task: Classify $x^2+4xy+y^2$ and find a direction of each sign.

Worked answer: $A=[[1,2],[2,1]]$ has eigenvalues 3 and -1, so it is indefinite. $q(1,1)=6$ and $q(1,-1)=-2$.

Principal axes and the Rayleigh quotient

A real symmetric matrix admits $A=Q\Lambda Q^T$ with Q orthogonal. Set $x=Qz$, equivalently $z=Q^Tx$. Orthogonal coordinates preserve lengths while removing cross terms: $x^TAx=z^T\Lambda z$. This is a change of coordinates, not a row reduction.

$\lambda\min \leq (x^TAx)/(x^Tx) \leq \lambda\max$ · $x=Qz \Rightarrow q=\Sigma\ \lambda_i z_i^2$

Work it through: For $A=[[2,1],[1,2]]$, normalized eigenvectors are $(1,-1)/\sqrt{2}$ and $(1,1)/\sqrt{2}$. In these coordinates $q=z_1^2+3z_2^2$. On $||x||=1$, the minimum is 1 and the maximum is 3. The ellipse $q=1$ has semiaxes 1 and $1/\sqrt{3}$.

Watch the trap: For a non-symmetric matrix, apply quadratic-form claims to its symmetric part; eigenvalues of the original matrix need not govern the quotient.

Recall task: For $A=\mathrm{diag}(2,7)$, bound $q(x)$ when $||x||=3$.

Worked answer: $18\leq q(x)\leq 63$, because $x^Tx=9$ and the eigenvalue bounds are 2 and 7.

Quadratic minimization and least-squares curvature

Consider $f(x)=\frac{1}{2}x^TAx-b^Tx+c$ with A symmetric. Its gradient is $Ax-b$ and its Hessian is A. These are multivariable supporting concepts; core single-variable differentiation is taught in the calculus recap. The linear algebra of solving $Ax=b$ already gives stationary points.

$\nabla f=Ax-b$ · $A>0 \Rightarrow x^*=\mathrm{solve}(A,b)$ · $\nabla^2\frac{1}{2}||Mx-y||^2=M^TM$

Work it through: Take $f(x,y)=x^2+2y^2-2x-4y$. $A=\mathrm{diag}(2,4)$, $b=(2,4)$, so $Ax=b$ gives $(1,1)$. Complete squares: $f=(x-1)^2+2(y-1)^2-3$. The unique minimum is -3 at $(1,1)$.

Watch the trap: A stationary point is not automatically a minimum. Check curvature and, for singular matrices, whether a solution exists.

Recall task: Minimize $\frac{1}{2}(3x^2+2y^2)-6x-4y$.

Worked answer: Solve $\mathrm{diag}(3,2)(x,y)=(6,4)$: $(2,2)$. The minimum is -10.

11 / LU, pivoting & stable solves

Elimination factors, triangular substitution and numerical judgment

Elimination becomes LU decomposition

Gaussian elimination subtracts multiples of pivot rows. Store these multipliers below the diagonal of a unit lower triangular L; the resulting upper triangular matrix is U. Without row swaps, the factorization is A=LU. It packages elimination so many right-hand sides can reuse the same work.

$A=LU \cdot l_ik=u_ik/u_kk \cdot Ly=b,\ Ux=y \cdot \det(A)=\Pi u_{ii}$ (unit L)

Work it through: A=[[2,1],[4,3]]. The first multiplier is 4/2=2. Row 2 becomes [0,1], so U=[[2,1],[0,1]]. L=[[1,0],[2,1]], and direct multiplication gives LU=A. For b=(3,7), Ly=b gives y=(3,1); Ux=y gives x=(1,1).

Watch the trap: L stores elimination multipliers with their original signs, because it reverses the row subtraction.

Recall task: Find unit-lower LU for [[3,2],[6,5]].

Worked answer: L=[[1,0],[2,1]], U=[[3,2],[0,1]].

Permutation matrices and partial pivoting

A permutation matrix reorders rows when multiplied on the left. A row swap changes determinant sign; every permutation matrix is orthogonal. With pivoting, this guide uses PA=LU. Always check the convention: some software reports A=PLU, where its P is the inverse of ours.

$PA=LU \Rightarrow Ly=Pb,\ Ux=y \cdot P^{-1}=P^T \cdot \det(A)=(-1)^s\ \Pi u_{ii}$

Work it through: A=[[0,2],[1,3]] has a zero first pivot but determinant −2. Swap rows: P=[[0,1],[1,0]], PA=[[1,3],[0,2]]. L=I and U=PA. For b=(4,7), Pb=(7,4). Back substitution gives y-coordinate 2, then x-coordinate 1.

Watch the trap: A zero first diagonal entry does not prove singularity; try a row interchange.

Recall task: Solve [[0,1],[2,3]]x=(2,8) by swapping rows first.

Worked answer: After swapping, $2x_1+3x_2=8$ and $x_2=2$, so x=(1,2).

Cholesky, conditioning and residuals

For a real symmetric positive definite matrix, Cholesky gives $A=LL^T$ with L lower triangular and positive diagonal. It uses symmetry and requires less work than general LU. Positive semidefinite matrices may require special handling; the standard positive-pivot algorithm assumes positive definiteness.

$A>0 \Rightarrow A=LL^T \cdot r=b-A\hat{x} \cdot \kappa_2(A)=\sigma max/\sigma min$

Work it through: For A=[[4,2],[2,3]], $l_{11}=2$ and $l_{21}=1$. $l_{22}=\sqrt{(3-1)}=\sqrt{2}$, so L=[[2,0],[1,√2]]. Multiply LL^T to recover A. For diag(1,0.001), $\kappa_2=1000$: small perturbations can be strongly amplified.

Watch the trap: A small determinant is scale-dependent; use singular values or a condition number to judge sensitivity.

Recall task: Find Cholesky L for [[9,3],[3,5]].

Worked answer: L=[[3,0],[1,2]], since the final diagonal is $\sqrt{(5-1)}=2$.

12 / SVD, pseudoinverse & low-rank models

Rectangular matrices, singular directions, minimum-norm solutions and compression

Singular value decomposition from first principles

Every real m×n matrix has $A=U\Sigma V^T$. In the full form, U is m×m orthogonal, V is n×n orthogonal, and Σ is m×n with nonnegative diagonal singular values sorted in descending order. Unlike eigendecomposition, SVD works for rectangular and defective matrices.

$$Av_i=\sigma_i u_i \cdot \sigma_i=\sqrt{\lambda_i(A^TA)} \cdot \operatorname{rank}(A)=\#\{\sigma_i>0\}$$

Work it through: Take A=[[3,0],[0,2],[0,0]], a 3×2 matrix. $A^TA=\operatorname{diag}(9,4)$, giving singular values 3 and 2. Right directions are e_1,e_2; corresponding left directions are the first two axes in R^3. The economy shapes are U:3×2, Σ:2×2, V^T:2×2.

Watch the trap: Singular values are nonnegative; they are not generally the eigenvalues or absolute diagonal entries of A.

Recall task: Find singular values and rank of [[0,2],[0,0]].

Worked answer: $A^TA=\operatorname{diag}(0,4)$; singular values are 2 and 0, so rank is 1. Both eigenvalues of A itself are 0.

Pseudoinverse and minimum-norm least squares

The Moore-Penrose pseudoinverse is $A^+=V\Sigma^+U^T$: invert each positive singular value and transpose the rectangular diagonal shape, leaving zeros at zero. It exists for every matrix, equals A^{-1} when an ordinary inverse exists, and can solve singular or rectangular systems.

$$A^+=V\Sigma^+U^T \cdot x_{LS}=A^+b \cdot AA^+=P_{col(A)} \cdot A^+A=P_{row(A)}$$

Work it through: Solve $x_1+x_2=2$, so A=[1,1] and b=2. The pseudoinverse is $[1/2,1/2]^T$, giving $x^*=(1,1)$. All solutions are $(1+t,1-t)$. Their squared norms are $2+2t^2$. The unique smallest norm occurs at t=0.

Watch the trap: A pseudoinverse returns a least-squares solution; it does not make an inconsistent system exact.

Recall task: Find the minimum-norm solution to $2x_1+2x_2=4$.

Worked answer: The solution is (1,1). It satisfies the equation and is perpendicular to null(A)=span((1,−1)).

Truncated SVD and the bridge to PCA

Write $A=\Sigma_i\sigma_i u_i v_i^T$ as a sum of rank-one outer products. Keeping the largest k singular values gives A_k, a best rank-at-most-k approximation in both spectral and Frobenius norms. This is the Eckart-Young result; uniqueness can fail when the cutoff singular values tie.

$$A_k=\Sigma_{i\le k}\,\sigma_i u_i v_i^T \cdot ||A-A_k||_F^2=\Sigma_{i>k}\,\sigma_i^2 \cdot \text{PCA variance}=\sigma_i^2/(n-1)$$

Work it through: Suppose singular values are 5,3,1. Total squared energy is 25+9+1=35. The best rank-one squared Frobenius error is 9+1=10. Rank two retains 34/35 of the energy, approximately 97.14%. Its spectral error is 1 and its Frobenius error is also 1.

Watch the trap: Energy uses squared singular values, not their sum. Center data before interpreting SVD directions as PCA axes.

Recall task: For singular values 4,2,1, find rank-one spectral error and squared Frobenius error.

Worked answer: Spectral error 2; squared Frobenius error $2^2+1^2=5$.

A final retrieval pass

1. A 4×3 matrix maps…

R^3 to R^4 / R^4 to R^3 / R^4 to R^4 only / R^3 to R^3 only

2. For basis (1,1),(1,−1), coordinates of (4,2) are…

(2,4) / (3,1) / (4,2) / (1,3)

3. A is 2×3 and B is 3×4. AB is…

2×4 / 3×3 / 4×2 / Undefined

4. The determinant of a real orthogonal matrix is…

Any nonzero real number / +1 or −1 / Always +1 / 0

5. Over R, a linear system can have…

Exactly three distinct solutions / Only one solution / Zero, one or infinitely many solutions / Exactly two distinct solutions

6. Full column rank means…

Every b has a solution regardless of shape / A must be square / A is orthogonal / Ax=0 has only x=0

7. For a 3×3 A with det A=2, det(3A) is…

18 / 27 / 54 / 6

8. Gram-Schmidt subtracts…

The whole previous input vector always / Only lengths / All coordinates / Projections on previous orthonormal directions

9. If Av=3v, then $(A^2-I)v$ equals…

8v / 2v / 9v / 0

10. Eigenvalues −2 and 3 make a symmetric form…

Positive definite / Negative definite / Positive semidefinite / Indefinite

Answers & reasoning

1. R^3 to R^4

Columns count input coordinates; rows count output coordinates.

2. (3,1)

Solve $a+b=4$ and $a-b=2$.

3. 2×4

The inner dimensions match and the outer dimensions remain.

4. +1 or −1

Take determinants in $Q^TQ=I$ to get $\det(Q)^2=1$.

5. Zero, one or infinitely many solutions

Two distinct solutions generate an affine line of solutions.

6. Ax=0 has only x=0

The nullity is zero.

7. 54

Scaling all three rows contributes $3^3\times2$.

8. Projections on previous orthonormal directions

Removing projected components leaves an orthogonal residual.

9. 8v

Evaluate the polynomial $3^2-1=8$.

10. Indefinite

Both signs occur along eigenvector directions.